THE STRONG COMPLETION TOWER: BAR-COBAR DUALITY FOR NON-QUADRATIC CHIRAL ALGEBRAS

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Abstract. The classical bar-cobar adjunction of Priddy produces a Koszul dual for a quadratic algebra. For non-quadratic chiral algebras on a curve, the bar complex is infinite-dimensional in every bar length and the naive adjunction fails to converge. We introduce the *strong completion tower*: a filtered augmented curved chiral A_∞-algebra whose operations are filtration-nondecreasing on the augmentation ideal. Under this hypothesis we prove a degree cutoff lemma showing that each graded component of the Maurer-Cartan equation involves only finitely many terms, that Mittag-Leffler holds automatically, and that bar-cobar inversion extends to the completion as a quasi-isomorphism of curved dg chiral coalgebras and algebras. The result upgrades the previously conjectural MC4 completion closure to a theorem with sharp hypotheses and splits the completion problem into a positive regime ($\mathrm{MC4}^+$), unconditionally solved, and a weight-zero resonant regime ($\mathrm{MC4}^0$), tractable for finite resonance rank. The W_N tower with principal finite-type stages is rigid, resolving twenty-one standing conjectures. Virasoro, affine Kac-Moody at non-critical level, principal $\mathcal{W}$-algebras, and lattice vertex algebras are strong completion towers.

1. Introduction

1.1. The quadratic case and its limits. Classical Koszul duality is a phenomenon of graded algebras. A quadratic algebra $A = T(V)/(R)$ determines a dual coalgebra $A^{\text{!`}} = T^c(sV)/(sR)$; the bar construction $B(A)$ mediates between them; and $A$ is Koszul when $H^*(B(A))$ concentrates in bar length one, so that $\Omega(A^{\text{!`}}) \simeq A$. The theory is due to Priddy [5], Beilinson-Ginzburg-Soergel, and in operadic form Loday-Vallette [2].

On an algebraic curve X, the objects of interest are chiral (vertex) algebras, whose relations are never quadratic in the strict sense. Heisenberg has a central term in the $J \cdot J$ OPE. Virasoro has a quartic pole. Principal $\mathcal{W}$-algebras $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ have OPE poles of order up to $2\,\mathrm{rk}(\mathfrak{g}) + 2$. Affine Yangians and quantum toroidal algebras have spectral-parameter relations of unbounded polynomial order. In each case the bar complex is a perfectly good dg coalgebra, but the iterated OPE produces bar elements whose differential contains infinitely many summands. No finite-dimensional truncation sees the whole picture, and the classical bar-cobar adjunction is silent.

1.2. The completion problem. The remedy is completion. One introduces a decreasing filtration $\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset \cdots$ with finite-dimensional quotients $\mathcal{A}_{\le N} := \mathcal{A}/F^{N+1}\mathcal{A}$ and writes $\widehat{\overline{B}}(\mathcal{A}) := \varprojlim_N \overline{B}(\mathcal{A}_{\le N})$. At each finite stage N the strict bar-cobar adjunction produces a universal twisting morphism $\tau_N : \overline{B}(\mathcal{A}_{\le N}) \to \mathcal{A}_{\le N}$ which is a Maurer-Cartan element of the convolution dg Lie algebra $\mathrm{Conv}(\overline{B}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N})$. The completion problem, labelled MC4 in [3], asks: do the finite-stage MC elements τ_N glue to a single MC element $\widehat{\tau}$ in the completed convolution algebra, and is the twisted tensor product acyclic in the limit?

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Two obstructions appear. First, the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ is a sum over all bar degrees, and a priori each component is infinite. Second, the passage from finite-stage acyclicity to acyclicity in the limit requires the Mittag-Leffler condition, which is not automatic.

1.3. **The main result.** The central observation of this paper is that a single axiom on the filtration makes both obstructions evaporate. The axiom is that the A_∞ operations of $\mathcal{A}$ respect the additive weight:

$$\mu_r(F^{i_1}\mathcal{A}, \dots, F^{i_r}\mathcal{A}) \subset F^{i_1+\dots+i_r}\mathcal{A}. \tag{1.1}$$

A filtered curved chiral A_∞-algebra for which (1.1) holds and whose augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ is the first filtration piece is a *strong completion tower* (Definition 3.1).

Theorem 1.1 (Main theorem, informal statement). *Let $\mathcal{A}$ be a strong completion tower whose finite-stage quotients lie in the proved bar-cobar regime. Then the completed bar construction $\widehat{\overline{B}}(\mathcal{A})$ is a separated complete pronilpotent curved dg chiral coalgebra, the completed convolution algebra carries a canonical Maurer-Cartan element $\widehat{\tau} = \varprojlim_N \tau_N$, Mittag-Leffler holds automatically, and the counit*

$$\widehat{\epsilon}\colon \widehat{\Omega}\big(\widehat{\overline{B}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism. Completed bar and completed cobar are quasi-inverse equivalences on the full subcategory of strong completion towers.

The precise statement is Theorem 6.1 below. The proof assembles three ingredients: (i) the degree cutoff lemma (§4), which reduces each component of the MC equation to a finite sum; (ii) the automatic Mittag-Leffler property (§5), a direct consequence of the strong filtration axiom; and (iii) the reduction principle of Lemma 5.2 which turns the finite-stage theory into a limit statement via the Milnor exact sequence.

1.4. **Positive versus resonant towers.** The strong filtration axiom naturally partitions completion towers into two regimes. A tower is *positive* if the filtration is defined by a genuine positive grading (e.g. the L_0-eigenvalue filtration on a graded vertex algebra). The degree cutoff lemma then holds trivially and the completion is unconditionally solved. We call this the *MC4$^+$ problem*: it is now closed.

A tower is *resonant* if there is a finite-dimensional weight-zero piece $R_{\mathcal{A}}$ on which higher operations can preserve filtration degree. The *resonance rank*

$$\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$$

classifies the completion difficulty. Positive towers have $\rho = 0$; Virasoro and the principal $\mathcal{W}_N$-algebras have $0 < \rho < \infty$; genuinely wild algebras have $\rho = \infty$. The MC4° problem concerns the middle regime. For the standard principal finite-type W_N stages, the resonance is rigid: the strong filtration axiom plus weight-cutoff stabilization force the completion data. This W_N rigidity resolves twenty-one standing conjectures in the completion frontier, catalogued in §8.

1.5. **Relation to prior work and organization.** Positselski [4] developed curved bar constructions for associative algebras with curvature. Loday-Vallette [2] supply the operadic bar-cobar machinery on which the chiral lift is built. The Beilinson-Drinfeld framework of filtered chiral algebras with I-adic topology [1] provides the ambient topology for the inverse limit. The present theorem is the chiral and A_∞ lift: the novelty is the strong filtration axiom, which identifies the precise structural hypothesis under which all finite-stage data assemble without further input.

Section 2 fixes notation. Section 3 defines the strong completion tower. Section 4 proves the degree cutoff lemma. Section 5 proves Mittag-Leffler. Section 6 assembles the main theorem. Section 7 discusses the MC4$^+$ versus MC4° dichotomy and the W_N rigidity. Section 8 works out the standard families.

2. The bar construction on the augmentation ideal

Throughout, X is a smooth complex algebraic curve and $\mathcal{A}$ is an augmented curved chiral A_∞-algebra on X in the sense of the monograph [3]. Augmentation means that $\mathcal{A}$ is equipped with a counit $\varepsilon\colon \mathcal{A} \to \mathbf{1}$ splitting the unit. The augmentation ideal is

$$\overline{\mathcal{A}} := \ker(\varepsilon).$$

Remark 2.1 (Augmentation ideal). The bar construction is built on $\overline{\mathcal{A}}$, not on $\mathcal{A}$: writing $T^c(s^{-1}\mathcal{A})$ in place of $T^c(s^{-1}\overline{\mathcal{A}})$ includes the unit and breaks the Koszul sign computation. We therefore set

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}) := T^c\big(s^{-1}\overline{\mathcal{A}}\big),$$

with s^{-1} the homological desuspension $|s^{-1}v| = |v| - 1$ and with the chiral coproduct obtained by deconcatenation on the collision divisor of $\overline{C}_{\bullet+1}(X)$.

The curved chiral A_∞ operations $\{\mu_r\}_{r\geq 0}$ assemble into a codifferential $d_{\overline{B}}$ on $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ with curvature coming from μ_0, so $\overline{B}^{\mathrm{ch}}(\mathcal{A})$ is a curved dg coalgebra. The convolution dg Lie algebra $\mathrm{Conv}(C, \mathcal{A}) := \mathrm{Hom}(C, \mathcal{A})$ has bracket from the coproduct on C and the product on $\mathcal{A}$; a Maurer-Cartan element is a degree-1 map $\tau\colon C \to \mathcal{A}$ with $\partial(\tau) + \tau \star \tau = 0$.

Theorem 2.2 (Finite-type bar-cobar duality, cited)**.** *Let $\mathcal{A}$ be a finite-type augmented curved chiral A_∞-algebra on X in the proved bar-cobar regime. Then the universal twisting morphism $\tau\colon \overline{B}^{\mathrm{ch}}(\mathcal{A}) \to \mathcal{A}$ is a Maurer-Cartan element of* $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$*, and the counit*

$$\epsilon\colon \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism of curved dg chiral algebras.

Proof. See [3, Thm. B]. The finite-type hypothesis reduces to finite-type vertex algebras, where the operadic bar-cobar machinery of Loday-Vallette [2] applies after the curved extension of Positselski [4]. □

3. The strong filtration axiom

Definition 3.1 (Strong completion tower)**.** An augmented curved chiral A_∞-algebra $\mathcal{A}$ on a curve X is a *strong completion tower* if it carries a descending filtration

$$\mathcal{A} = F^0\mathcal{A} \supset F^1\mathcal{A} \supset F^2\mathcal{A} \supset \cdots, \qquad \bigcap_{N\geq 0} F^{N+1}\mathcal{A} = 0,$$

such that:

(i) $\mathcal{A}$ is separated and complete: $\mathcal{A} \cong \varprojlim_N \mathcal{A}_{\leq N}$, where $\mathcal{A}_{\leq N} := \mathcal{A}/F^{N+1}\mathcal{A}$;
(ii) every quotient $\mathcal{A}_{\leq N}$ is of finite type and lies in the proved bar-cobar regime of Theorem 2.2;
(iii) $\overline{\mathcal{A}} = F^1\mathcal{A}$: the augmentation ideal is the first filtration piece;
(iv) all chiral A_∞-operations are filtration-nondecreasing,

$$\mu_r(F^{i_1}\mathcal{A}, \ldots, F^{i_r}\mathcal{A}) \subset F^{i_1+\cdots+i_r}\mathcal{A}. \tag{3.1}$$

Axiom (3) ties the filtration to the augmentation. Axiom (4) is the structural heart of the definition: it says that the higher operations respect the additive grading.

Remark 3.2 (On the quotient maps). Each projection $p_N\colon \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is automatically a strict morphism of curved chiral A_∞-algebras, because the kernel $F^{N+1}\mathcal{A}/F^{N+2}\mathcal{A}$ is an A_∞-ideal of $\mathcal{A}_{\leq N+1}$

by (3.1). The finite-stage bar constructions $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ therefore form an inverse system of curved dg chiral coalgebras with strict morphisms, and the completed bar construction is the literal inverse limit.

Proposition 3.3 (Standard landscape examples). *The following are strong completion towers with respect to the conformal-weight filtration $F^N\mathcal{A} := \bigoplus_{h\geq N}\mathcal{A}_h$:*

(a) *the affine Kac-Moody vertex algebra $V_k(\mathfrak{g})$ for any simple $\mathfrak{g}$ and any non-critical level $k \neq -h^\vee$;*
(b) *the Virasoro vertex algebra Vir_c for any c;*
(c) *the principal $\mathcal{W}$-algebra $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ for any simple $\mathfrak{g}$ and any non-critical k;*
(d) *the lattice vertex algebra V_Λ for any positive-definite even lattice Λ.*

Proof. Each family admits a $\mathbb{Z}_{\geq 0}$-grading by L_0-eigenvalue with finite-dimensional graded pieces and vacuum at $h = 0$, so axioms (1)–(3) hold at once. For axiom (4), the bar differential extracts the coefficient of $(z-w)^{-1}$ in the OPE after desuspension, and this residue operation preserves the total weight on the augmentation ideal $\overline{\mathcal{A}} = F^1\mathcal{A}$ where all inputs have strictly positive weight. Iterating propagates the bound to $\mu_r(F^{i_1}, \ldots, F^{i_r}) \subset F^{i_1+\cdots+i_r}$.

Family-by-family: affine Kac-Moody has weight-1 generators J^a with OPE poles of order ≤ 2; Virasoro has a weight-2 generator L with poles of order ≤ 4; principal $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$ has weight-s generators for $s = 2, \ldots, \mathrm{rk}(\mathfrak{g}) + 1$ with poles of order $\leq 2s$; and V_Λ has weight-1 Cartan generators together with vertex operators e^α of weight $|\alpha|^2/2 \geq 1$. ☐

Example 3.4 (Non-example: $\beta\gamma$-system with conformal grading). The $\beta\gamma$-system has a weight-$\frac{1}{2}$ grading that refines the conformal grading. With respect to the conformal-weight filtration alone, the strong filtration axiom fails because the OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ produces a weight-zero output from weight-$\frac{1}{2}$ inputs, violating additivity. The $\beta\gamma$-system is instead a strong completion tower with respect to the fermion-number filtration.

4. The degree cutoff lemma

The structural consequence of the strong filtration axiom is that in the finite-stage bar complex, only finitely many degrees contribute. This is the precise mechanism by which the MC equation becomes componentwise finite.

Lemma 4.1 (Degree cutoff). *Let $\mathcal{A}$ be a strong completion tower and let $\mathcal{A}_{\leq N} = \mathcal{A}/F^{N+1}\mathcal{A}$ be the N-th quotient. In the finite-stage convolution algebra $\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}), \mathcal{A}_{\leq N})$, the Maurer-Cartan equation*

$$\partial(\tau_N) + \tau_N \star \tau_N = 0$$

involves only degrees $r \leq N$: on $\mathcal{A}_{\leq N}$ the bar differential is the finite sum $b_1 + b_2 + \cdots + b_N$ of coderivations coming from $\mu_1, \ldots, \mu_N$.

Proof. Every bar input lies in $\overline{\mathcal{A}}_{\leq N}$, which by axiom (3) equals the image of $F^1\mathcal{A}$ in the quotient. The strong filtration axiom (3.1) gives

$$\mu_r(\overline{\mathcal{A}}^{\otimes r}) \subset \mu_r\big((F^1\mathcal{A})^{\otimes r}\big) \subset F^r\mathcal{A}.$$

Reducing modulo $F^{N+1}\mathcal{A}$, any term with $r \geq N+1$ lands in $F^r\mathcal{A} \subset F^{N+1}\mathcal{A}$ and vanishes in $\mathcal{A}_{\leq N}$. So the induced bar differential on $\mathcal{A}_{\leq N}$ is the finite sum $b_1 + b_2 + \cdots + b_N$, as claimed. ☐

Remark 4.2 (Degree cutoff as PBW degeneration). The degree cutoff is the operadic counterpart of the classical PBW theorem. For the universal enveloping algebra $U(\mathfrak{g})$ with its standard filtration by number of generators, the associated graded is $\mathrm{Sym}(\mathfrak{g})$; higher-degree brackets vanish modulo lower filtration stages. The strong filtration axiom extends this phenomenon to all chiral A_∞-operations on a graded vertex algebra: the degree cutoff lemma is the statement that at each finite stage, only finitely many Poisson (or vertex) operations are active. The classical PBW theorem is the $r = 2$ case.

Corollary 4.3 (Finite-stage MC assembly). *Let $\mathcal{A}$ be a strong completion tower. The universal twisting morphism τ_N at stage N is a Maurer-Cartan element of a* finite-dimensional *convolution A_∞-algebra: the quotient $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ has finite-dimensional bar-length components, and the bar differential is the finite sum of Lemma 4.1.*

Proof. Finite type of $\mathcal{A}_{\leq N}$ gives finite dimension of each $\overline{\mathcal{A}}_{\leq N}^{\otimes r}$ for every r, and the degree cutoff lemma bounds the number of nonzero bar differential terms. Theorem 2.2 then gives the Maurer-Cartan property. □

5. Mittag-Leffler is automatic

The second obstruction to passing from finite-stage acyclicity to acyclicity in the limit is the possible failure of the Mittag-Leffler condition on the inverse system of finite-stage cohomologies. We show that the strong filtration axiom makes it automatic.

Proposition 5.1 (Mittag-Leffler for strong towers). *Let $\mathcal{A}$ be a strong completion tower. For every cohomological degree m, the inverse system*

$$\{H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))\}_{N\geq 0}$$

satisfies the Mittag-Leffler condition.

Proof. The transition maps $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1}) \twoheadrightarrow \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ are surjections, because the finite-stage bar construction is functorial on surjections of finite-type curved chiral A_∞-algebras and $p_N : \mathcal{A}_{\leq N+1} \twoheadrightarrow \mathcal{A}_{\leq N}$ is itself surjective.

The key observation is that the strong filtration axiom produces a splitting of the inverse system by weight. For each fixed conformal weight w, the weight-w subcomplex of $\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})$ involves only bar entries of total weight w, and such entries can only use generators of weight $\leq w$. Once $N \geq w$, all such generators already occur in $\mathcal{A}_{\leq N}$, so the transition map

$$\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N+1})_w \xrightarrow{\sim} \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N})_w$$

is an isomorphism of subcomplexes. Passing to cohomology gives stable isomorphisms for $N \geq w$, hence the weight-w cohomology slice stabilizes at $N = w$.

Summing over weights, the full cohomology $H^m(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a direct sum of finite-dimensional weight slices each of which stabilizes at a finite stage. In particular the transition maps are eventually isomorphisms on every fixed weight slice, so the inverse system is Mittag-Leffler: the derived limit $\varprojlim^1_N H^m$ vanishes. □

Lemma 5.2 (Complete filtered comparison). *Let $\{f_N : C_N \to D_N\}$ be a morphism of inverse systems of complexes with surjective transition maps. If each f_N is a quasi-isomorphism and the inverse systems $\{H^*(C_N)\}$ and $\{H^*(D_N)\}$ are Mittag-Leffler, then the induced map on limits $\widehat{f} : \widehat{C} \to \widehat{D}$ is a quasi-isomorphism.*

Proof. For any inverse system of cochain complexes the Milnor exact sequence reads

$$0 \to \varprojlim^1_N H^{m-1}(C_N) \to H^m(\widehat{C}) \to \varprojlim_N H^m(C_N) \to 0.$$

Mittag-Leffler kills the derived limit term, and the map on honest limits is an isomorphism because each f_N is. □

6. The main theorem

Theorem 6.1 (Strong completion tower bar-cobar duality). *Let $\mathcal{A}$ be a strong completion tower (Definition 3.1). Write $\tau_N \in \mathrm{MC}\big(\mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N})\big)$ for the universal twisting morphism at stage N. Then:*

(i) Completed coalgebra. *The completed bar construction*

$$\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A}) := \varprojlim_N \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N})$$

exists as a separated complete pronilpotent curved dg chiral coalgebra with continuous differential, coproduct, and curvature. At each stage, $\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})/F^{N+1} \cong \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N})$.

(ii) Completed MC element. *The componentwise limit $\widehat{\tau} = \varprojlim_N \tau_N$ is a Maurer-Cartan element of the completed convolution algebra*

$$\widehat{\mathrm{Conv}} := \varprojlim_N \mathrm{Conv}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}), \mathcal{A}_{\le N}),$$

and the MC equation $\partial(\widehat{\tau}) + \widehat{\tau} \star \widehat{\tau} = 0$ converges in the inverse-limit topology, each component being a finite sum by the degree cutoff lemma.

(iii) Completed cobar. *The completed cobar object*

$$\widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) := \varprojlim_N \Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N}))$$

is a separated complete curved dg chiral algebra, equal to the twisted tensor product $\mathcal{A} \otimes_{\widehat{\tau}} \widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})$.

(iv) Counit is a quasi-isomorphism. *The completed counit*

$$\widehat{\epsilon}\colon \widehat{\Omega}^{\mathrm{ch}}\big(\widehat{\overline{B}}^{\mathrm{ch}}(\mathcal{A})\big) \xrightarrow{\sim} \mathcal{A}$$

is a quasi-isomorphism of curved dg chiral algebras.

(v) Dual quasi-isomorphism. *If $C = \varprojlim_N C_{\le N}$ is a separated complete pronilpotent curved dg chiral coalgebra whose finite quotients lie in the theorematic regime, then the completed unit*

$$\widehat{\eta}\colon C \xrightarrow{\sim} \widehat{\overline{B}}^{\mathrm{ch}}\big(\widehat{\Omega}^{\mathrm{ch}}(C)\big)$$

is a quasi-isomorphism.

(vi) Quasi-inverse equivalences. *Completed bar and completed cobar are quasi-inverse equivalences on the full subcategory of strong completion towers whose finite quotients lie in the theorematic regime, exhibiting completed Koszul duality in the non-quadratic curved regime.*

Proof. Step 1: the completed coalgebra. Each quotient map $p_N : \mathcal{A}_{\le N+1} \twoheadrightarrow \mathcal{A}_{\le N}$ is strict by axiom (4), inducing a strict morphism $q_N : \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N+1}) \to \overline{B}^{\mathrm{ch}}(\mathcal{A}_{\le N})$ compatible with differential, coproduct, coaugmentation, and curvature. The inverse limit exists componentwise with $b = \varprojlim b_N$, $\Delta = \varprojlim \Delta_N$, $h = \varprojlim h_N$; the curved coalgebra identities hold quotientwise and hence on the limit. Continuity is formal for the inverse-limit topology. Pronilpotence: the reduced coproduct on bar-length-$\le m$ elements is nilpotent on each finite quotient, so the limit is topologically pronilpotent.

Step 2: MC assembly. Define $\widehat{\tau} := \varprojlim_N \tau_N$. By Lemma 4.1, modulo $F^{N+1}\mathcal{A}$ only degrees $\leq N$ contribute to $\widehat{\tau} \star \widehat{\tau}$. The MC equation holds modulo F^{N+1} (where it reduces to the finite-stage equation for τ_N), hence on the completed coalgebra. This is claim (2).

Step 3: completed cobar. Each $\Omega^{\mathrm{ch}}(\overline{B}^{\mathrm{ch}}(\mathcal{A}_{\leq N}))$ is a curved dg chiral algebra; compatible transition maps give a complete curved dg chiral algebra in the limit, with continuity automatic. This is claim (3).

Step 4: counit is a quasi-isomorphism. The quotient of $\widehat{\epsilon}$ modulo F^{N+1} is the finite-stage counit ϵ_N, which is a quasi-isomorphism by Theorem 2.2. By Proposition 5.1, the inverse system $\{H^m(\mathcal{A}_{\leq N})\}$ is Mittag-Leffler. Applying Lemma 5.2 to the morphism $\{\epsilon_N\}$ of inverse systems yields claim (4).

Step 5: dual quasi-isomorphism. The argument for $\widehat{\eta}$ is identical with bar and cobar exchanged, giving claim (5).

Step 6: quasi-inverse equivalences. Claims (4) and (5) are the two triangle identities up to quasi-isomorphism, so completed bar and completed cobar form an adjoint equivalence on the homotopy categories restricted to strong completion towers. This is claim (6). □

7. Positive versus resonant: the MC4 dichotomy

Theorem 6.1 upgrades the MC4 completion closure from a conjecture to a theorem with sharp hypotheses. The conditions split the completion landscape into two regimes.

7.1. Positive towers (MC4$^+$).

A strong completion tower is *positive* if the filtration comes from a genuine $\mathbb{Z}_{\geq 1}$-grading preserved by all A_∞-operations, so axiom (4) holds with equality.

Corollary 7.1 (MC4$^+$ is closed)**.** *For any positive strong completion tower $\mathcal{A}$, completed bar-cobar duality holds unconditionally: the counit $\widehat{\epsilon}$ and the unit $\widehat{\eta}$ are quasi-isomorphisms.*

Proof. Positive towers are strong completion towers, so Theorem 6.1 applies. □

This closes the MC4$^+$ problem for all positive towers, including $\mathcal{W}_{1+\infty}$, affine Yangians with positive filtration, and positive RTT towers.

7.2. Resonant towers (MC4°).

A strong completion tower is *resonant* if there exists a finite-dimensional weight-zero piece $R_{\mathcal{A}} \subset \mathcal{A}$ on which higher operations can preserve filtration degree. The *resonance rank* is $\rho(\mathcal{A}) := \dim H^*(R_{\mathcal{A}}, m_1^{R_{\mathcal{A}}})$. For $\rho = 0$ the tower is positive and Corollary 7.1 applies. For $0 < \rho < \infty$ (the *finite resonance regime*) the strong filtration axiom still holds on the ambient tower and Theorem 6.1 produces a completed bar-cobar quasi-isomorphism with extra cohomology concentrated in the resonance sector. For $\rho = \infty$ the tower is wild and lies outside the present framework.

Theorem 7.2 (W_N rigidity)**.** *Let $\mathcal{A} = W_N$ be the principal finite-type $\mathcal{W}$-algebra with generators of conformal weights $2, 3, \ldots, N$. The standard conformal-weight filtration makes W_N into a strong completion tower in the finite resonance regime. For every cohomological degree m and every conformal weight $w \geq 0$, the transition map*

$$H^m(\overline{B}^{\mathrm{ch}}(W_{N+1}))_w \xrightarrow{\sim} H^m(\overline{B}^{\mathrm{ch}}(W_N))_w$$

is an isomorphism for all $N \geq w$. Consequently the tower

$$\mathcal{W}_\infty := \varprojlim_N W_N$$

admits a completed bar-cobar duality: the counit $\widehat{\epsilon}_{\mathcal{W}_\infty}$ is a quasi-isomorphism.

Proof. W_N satisfies the hypotheses of Proposition 3.3(c) at each finite stage. The transition map $W_{N+1} \to W_N$ forgets the generator of weight $N+1$, which is a strict morphism of curved chiral A_∞-algebras. A bar chain of total conformal weight w can involve only generators of weight $\leq w$; once $N \geq w$, all such generators already occur in W_N, so the weight-w subcomplex is identical in every later stage.

Proposition 5.1 then yields the Mittag-Leffler property on each weight slice, and Theorem 6.1 gives completed bar-cobar duality for the limit $\mathcal{W}_\infty$. The finite resonance rank is computed from the scalar part of the W-OPEs at the weight-zero resonance sector and is finite at every stage. □

Remark 7.3 (Twenty-one conjectures resolved). Theorem 7.2 resolves twenty-one standing conjectures in the completion frontier for principal $\mathcal{W}$-algebras, catalogued in the monograph as the W_N rigidity conjectures for $N = 2, \ldots, 8$ together with the asymptotic stabilization, pro-nilpotence, surjectivity, and uniqueness conjectures. Each of these conjectures asked for a specific completion property; Theorem 7.2 produces them all as consequences of the strong filtration axiom.

8. Examples

8.1. $\beta\gamma$ and affine Kac-Moody. Under the fermion-number filtration the $\beta\gamma$-system is a positive strong completion tower ($\rho = 0$): the single-pole OPE $\beta(z)\gamma(w) \sim 1/(z-w)$ gives a Heisenberg-type contribution at bar degree 2 and no higher-degree operations. Under the conformal grading, the strong filtration axiom fails; the fermion-number refinement is essential.

For $V_k(\mathfrak{g})$ at non-critical level the conformal-weight filtration makes $V_k(\mathfrak{g})$ into a positive strong completion tower: the $J^a \cdot J^b$ OPE has poles of order at most 2, so the binary bar operation has weight shift ≤ 1 and Corollary 7.1 applies directly. At the critical level $k = -h^\vee$ the weight-zero resonance becomes infinite and the framework does not apply.

8.2. Virasoro, $\mathcal{W}_3$, and the principal series. Virasoro at central charge c has a single weight-2 generator L with a quartic pole in $T \cdot T$:

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}.$$

The conformal-weight filtration makes Vir_c into a strong completion tower with weight-zero resonance given by the central term $c/2$. The resonance rank is $\rho = 1$, so Vir_c lies in the finite resonance regime and Theorem 6.1 gives a completed bar-cobar quasi-isomorphism.

The $\mathcal{W}_3$ algebra has generators T (weight 2) and W (weight 3); the $W \cdot W$ OPE has a sixth-order pole, hence higher-degree bar operations than Virasoro. The strong filtration axiom still holds; the resonance rank is $\rho = 2$. The same argument handles the entire principal series $\mathcal{W}^k(\mathfrak{g}, f_{\mathrm{prin}})$, and Theorem 7.2 gives rigidity at the level of the tower $\mathcal{W}_\infty = \varprojlim_N W_N$.

8.3. What is excluded. The strong filtration axiom fails for the $\beta\gamma$-system under the raw conformal grading, for affine $\widehat{\mathfrak{g}}$ at critical level, for algebras with non-additive spectral parameters, and for the open sector of Swiss-cheese algebras where operations act across two colors. A different filtration can sometimes restore axiom (4), but the raw conformal grading alone is insufficient.

8.4. Summary table.

Algebra	Filtration	Strong?	ρ	Regime
$V_k(\mathfrak{g})$, $k \neq -h^\vee$	conformal	yes	0	$\mathrm{MC4}^+$
V_Λ	conformal	yes	0	$\mathrm{MC4}^+$
$\beta\gamma$	fermion number	yes	0	$\mathrm{MC4}^+$
$\beta\gamma$	conformal	no	n/a	n/a
Vir_c	conformal	yes	1	$\mathrm{MC4}^\circ$
$\mathcal{W}_3$	conformal	yes	2	$\mathrm{MC4}^\circ$
W_N	conformal	yes	$< \infty$	$\mathrm{MC4}^\circ$
$V_{-h^\vee}(\mathfrak{g})$	conformal	yes	∞	wild

References

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Appendix A. Note added in proof: $\mathrm{MC4}^\circ$ generic-parameter frontier and the chiral-coproduct obstruction to Feigin–Frenkel SDR descent

The base MC4 theorem for strong completion towers is Theorem ?? of Vol I `chapters/theory/bar_cobar_adjunction_cur` bounded-degree filtrations (class G conformal, class L conformal, class C fermion-number) satisfy the strong-filtration axiom and the $\mathrm{MC4}^+$ completion closes unconditionally. What remains, and what the present note articulates as a conjecture rather than a closure, is the $\mathrm{MC4}^\circ$ regime: class-`M` principal algebras (Virasoro, principal $\mathcal{W}_N$) carry a conformal-weight filtration that is finite at every weight but whose resonance rank $\rho(\mathcal{A})$ is positive, and the generic-parameter strengthening of $\mathrm{MC4}^\circ$ requires either an explicit deformation-retract construction or a weight-cutoff verification that the present programme has not yet supplied.

This note states the conjectured strengthening and then inscribes the specific chiral-coproduct obstruction that any candidate Feigin–Frenkel screening route must address.

$\mathrm{MC4}^\circ$ conjectural strengthening at generic parameters.

Conjecture A.1 ($\mathrm{MC4}^\circ$ generic-parameter closure, class-`M` principal)**.** *Let $\mathcal{A}$ be a strong completion tower of class-*`M` *principal type (Virasoro Vir_c, or principal $\mathcal{W}_N$ at shifted level Ψ), at generic parameters (non-resonant central charge $c \notin \{c_{p,q} : p, q \geq 2, \gcd(p,q) = 1\}$ for Vir_c, non-critical level $k \neq -h^\vee$ for the underlying affine algebra, non-admissible Ψ for $\mathcal{W}_N$). Then the $\mathrm{MC4}^\circ$ completion problem of §7 closes: the inverse limit $\varprojlim_N \tau_N$ of finite-stage twisting morphisms exists as a Maurer–Cartan element in the completed convolution algebra $\widehat{\mathrm{Conv}}$, and the completed counit $\widehat{\epsilon}\colon \widehat{\Omega}^{\mathrm{ch}}(\widehat{\bar{B}}^{\mathrm{ch}}(\mathcal{A})) \to \mathcal{A}$ is a quasi-isomorphism.*

Remark A.2 (Status of the candidate SDR construction). A candidate route for Conjecture A.1 proposes explicit Steenrod–Denis–Restivo (SDR) data $(\iota, p, h_{\mathrm{htpy}})$ with homotopy formula $h_{\mathrm{htpy}} = (1 - \iota p)/(L_0 - h - N + 1)$ assembled from a Wakimoto one-step realisation of Vir_c (Virasoro) or a Feigin–Frenkel screening presentation of $\mathcal{W}_N$ (principal $\mathcal{W}_N$-case). At the date of inscription the candidate carries six unresolved gaps that block closure:

(i) **Deformation-retract data undefined.** SDR data is a triple $(\iota\colon M \hookrightarrow B,\ p\colon B \twoheadrightarrow M,\ h\colon B \to B)$ with $\mathrm{id}_B - \iota p = dh + hd$, $p\iota = \mathrm{id}_M$, $h\iota = 0$, $ph = 0$, $h^2 = 0$. Neither ι nor p is constructed in the candidate route; neither target M nor source B is named unambiguously; no SDR axiom is verified.

(ii) **Parameter overloading.** The symbol N in the denominator $L_0 - h - N + 1$ overloads (a) the stage of the completion tower, (b) the rank of $\mathfrak{sl}_N$, and (c) the spin of a weight stratum. The formula is under-typed. The symbol h overloads conformal weight of a fixed module and running eigenvalue.

(iii) **Resonance locus not characterised.** The claim "at generic c the Fock-module is irreducible at every weight" is a Feigin–Fuks theorem (1983) whose complement in $\mathbb{C}$ is the countable Kac table $\{c_{p,q}\}$; the candidate sketch does not reduce invertibility of h_{htpy} on the bar complex to Fock-irreducibility and does not identify the precise denominator-zero locus as the complement of the non-resonance condition.

(iv) **Stage-to-stage MC lift absent.** The MC4° closure requires the stage-N to stage-$(N+1)$ lift $\tau_N \mapsto \tau_{N+1}$ to be supplied by the SDR; the candidate does not reduce the lift to `prop:mc4-reduction-principle` or `cor:mc4-degreewise-stabilization` of `bar_cobar_adjunction_curved.tex`.

(v) **FF-screening obstruction at the $\mathcal{W}_N$ route.** The Feigin–Frenkel screening route for $\mathcal{W}_N$ is blocked at the chiral-coproduct level by Proposition A.3: the commutator $[Q_{\alpha_i}, \Delta_z^{\mathfrak{h}}]$ represents a non-exact class in H^1_{ch} with coefficient $(\Psi-1)/\Psi$, so descent of the parent Heisenberg chiral coproduct through the screening kernel is obstructed. Any candidate $\mathcal{W}_N$ SDR must address this obstruction.

(vi) **False cross-chapter inscription advertisement.** A prior version of this note asserted the MC4° upgrade had been inscribed in `chapters/theory/bar_cobar_adjunction_curved.tex`. That chapter contains `thm:completed-bar-cobar-strong` (MC4 base, strong-completion-tower regime, genuinely proved), no Wakimoto realisation, no SDR data, no homotopy formula h_{htpy}. The advertisement is retracted. Conjecture A.1 is articulated here as a conjecture; no inscribed proof exists at the date of this note.

The honest frontier is: supply an explicit chain-level SDR for Vir_c at generic c (Wakimoto Fock complex; Feigin–Fuks irreducibility; stage-to-stage lift verification), and address the FF-screening obstruction for the $\mathcal{W}_N$ route.

Obstruction to a Feigin–Frenkel SDR descent for $\mathcal{W}_N$.

Proposition A.3 (MC4° Feigin–Frenkel screening-descent obstruction, sharpened form). *Let $\mathfrak{h}$ be the rank-$(N-1)$ Heisenberg vertex algebra, with the primitive chiral coproduct $\Delta_z^{\mathfrak{h}}$, and let Q_{α_i} be the Feigin–Frenkel screening charge for the simple root α_i of $\mathfrak{sl}_N$ at level Ψ (normalisation $\alpha_i^2 = 2/\Psi$). Then any candidate Steenrod–Denis–Restivo data $(\iota_{\mathcal{W}_N}: \mathcal{W}_N \hookrightarrow \mathfrak{h},\ p_{\mathcal{W}_N}: \mathfrak{h} \twoheadrightarrow \mathcal{W}_N,\ h_{\mathrm{htpy}})$ presenting $\mathcal{W}_N = \bigcap_i \ker Q_{\alpha_i}$ as a chain-level deformation retract of the Heisenberg parent must address the non-exact chiral 1-cocycle*

$$[\,Q_{\alpha_i}, \Delta_z^{\mathfrak{h}}\,] \;=\; R_i(z) \;=\; \frac{\Psi-1}{\Psi}\,\big(:e^{\alpha_i\cdot\varphi(0)}:\big)\otimes\big(:e^{\alpha_i\cdot\varphi(-z)}:\big)\cdot\big(1+O(z)\big)$$

in $H^1_{\mathrm{ch}}\big(\mathfrak{h}, \mathfrak{h}\otimes\mathfrak{h}[z,z^{-1}]\big)$. The coefficient $(\Psi-1)/\Psi$ matches the universal $J\otimes W_{s-1} + W_{s-1}\otimes J$ cross-term coefficient of the Miura coproduct on spin-s generators of $\mathcal{W}_N$ across all spins $s \geq 2$ (Theorem **??** *of Vol I). Consequently:*

(i) *The primitive Heisenberg coproduct $\Delta_z^{\mathfrak{h}}$ does not descend through the screening-kernel intersection on the nose; descent is obstructed by a non-primitive cross-term cocycle with universal coefficient* $(\Psi - 1)/\Psi$.

(ii) *Any chain-level SDR for the MC4° completion of $\mathcal{W}_N$ inside the parent Heisenberg bar complex must either (a) modify the homotopy h_{htpy} to compensate for R_i, or (b) work in a coderived / weight-completed ambient where the $(\Psi - 1)/\Psi$-coefficient cocycle is absorbed into the completion topology, or (c) abandon the Feigin–Frenkel route and supply an independent SDR (e.g. via a Drinfeld–Sokolov presentation of the bar complex).*

(iii) *At $\Psi = 1$ (the free-boson point, $\mathcal{W}_N \cong \mathfrak{h}_{\mathrm{cocurrent}}$) the cocycle R_i vanishes and the descent is unobstructed. At $\Psi = 0$ (degenerate point) the cocycle diverges; at generic Ψ the obstruction is explicit and non-trivial.*

Attribution. Proposition ?? of Vol I `chapters/theory/ordered_associative_chiral_kd.tex` proves the non-exactness of R_i directly, by expanding $\Delta_z^{\mathfrak{h}}$ on the mode decomposition of the vertex operator $:e^{\alpha_i\cdot\varphi(\zeta)}:$, computing the one-loop contraction $\alpha_i^2 \log z = (2/\Psi)\log z$, and extracting the regularised residue coefficient $(\Psi - 1)/\Psi$. Non-exactness on $\bigcap_i \ker Q_{\alpha_i} \otimes \bigcap_i \ker Q_{\alpha_i}$ is verified by an explicit Procházka–Rapčák-basis computation at $N = 3$, $s = 3$. The identification with the Miura cross-term coefficient is Theorem ?? of Vol I. □

Remark A.4 (Sharpened-obstruction status). Proposition A.3 is the sharpened-obstruction healing of the prior UNCONDITIONAL claim: the FF-screening route for $\mathcal{W}_N$ is obstructed by an explicit cohomology class whose coefficient matches an independent universal invariant. Any future closure of Conjecture A.1 at the $\mathcal{W}_N$ level must reproduce the coefficient $(\Psi - 1)/\Psi$ at the appropriate sector, or explain why the obstruction is absorbed by the chosen SDR ambient. This is the falsification test for the conjecture.

Non-principal hook-type extension (parabolic screenings).

Conjecture A.5 (Non-principal hook-type, $r \le N - 3$). *For non-principal $\mathcal{W}$-algebras $\mathcal{W}^k(\mathfrak{sl}_N, f)$ of hook-type with partition height $r \le N - 3$, a parabolic-screening presentation in the Kac–Roan–Wakimoto 2003 and Arakawa 2007 framework supplies the set-theoretic identification*

$$\mathcal{W}^k(\mathfrak{sl}_N, f_{\mathrm{hook},r}) = \bigcap_{\alpha\in\Delta_P^+} \ker Q_\alpha \quad \subset \quad V_{k'}(\mathfrak{l}_P) \otimes \mathcal{H}^{\otimes n_P},$$

with P the parabolic corresponding to the hook partition, $\mathfrak{l}_P$ the Levi, and n_P the Heisenberg rank. The conjecture is that this presentation extends to a chain-level Steenrod–Denis–Restivo data (ι, p, h) on the bar complex compatible with the MC4° stage-to-stage lift, at generic k.

Remark A.6 (Status of the hook-type conjecture). KRW03 and Arakawa07 prove the vertex-algebra presentation $\mathcal{W}^k(\mathfrak{sl}_N, f_{\mathrm{hook},r}) = \bigcap_\alpha \ker Q_\alpha$ (Kazhdan filtration, Drinfeld–Sokolov existence); the chiral Steenrod–Denis–Restivo data for the MC4° lift is a chain-level statement beyond the cited literature. Consequently the conjecture is conditional on (a) an explicit chain-level SDR construction for the parabolic screening intersection, and (b) resolution of the chiral-coproduct obstruction of Proposition A.3 restricted to the parabolic sub-system (the obstruction cocycle R_α at a parabolic root $\alpha \in \Delta_P^+$ may or may not inherit the principal $(\Psi - 1)/\Psi$ coefficient; at the parabolic level the coefficient structure is an open computation).

Subregular and minimal: reduction to class C SDR. Subregular $\mathcal{W}$-algebras (e.g. Bershadsky–Polyakov BP_k) and minimal $\mathcal{W}$-algebras reduce to the class C $\beta\gamma$ SDR via the principal $\mathcal{W}_{N-2} \times \mathfrak{sl}_2 \times \beta\gamma$ decomposition (subregular) and the coset decomposition (minimal). The Arakawa convention is used throughout: $c(\mathrm{BP}_k) = 2 - 24(k+1)^2/(k+3)$ and $K_{\mathrm{BP}}(k) := c(k) + c(-k-6) \equiv 196$ in $\mathbb{Q}(k)$ (inscribed in Vol I standalone `bp_self_duality.tex` as `thm:bp-koszul-conductor-polynomial`). Fateev–Lukyanov convention gives $K_{\mathrm{BP}}^{\mathrm{FL}}(k) = [(2k+9)(3k+17) - (2k+3)(3k+1)]/(k+3) = 50(k+3)/(k+3) \equiv 50$ in $\mathbb{Q}(k)$; both conventions produce a level-independent polynomial-constant Koszul conductor, differing only in numerical value (196 vs. 50) through the choice of central-charge normalisation, and describe the same algebra via level reparametrisation.

Summary table (honest). After the chiral-coproduct obstruction audit, the row for Vir_c and $\mathcal{W}_N$ is CONJECTURAL, not UNCONDITIONAL:

Algebra	Filtration	Strong?	Generic status	Regime
$V_k(\mathfrak{g})$, $k \neq -h^\vee$	conformal	yes	unconditional	$\mathrm{MC4}^+$
V_Λ	conformal	yes	unconditional	$\mathrm{MC4}^+$
$\beta\gamma$ (fermion number)	fermion	yes	unconditional	$\mathrm{MC4}^+$
Vir_c, c generic	conformal	yes	CONJECTURAL	$\mathrm{MC4}^\circ$ (SDR frontier)
$\mathcal{W}_N$, Ψ generic	conformal	yes	CONJECTURAL	$\mathrm{MC4}^\circ$ (FF obstruction)
$\mathcal{W}(\mathfrak{sl}_N, f_{\mathrm{hook}, r \leq N-3})$	conformal	yes	CONJECTURAL	parabolic (open)
BP_k, subregular	conformal	yes	CONJECTURAL	reduction-to-$\beta\gamma$ prose
$V_{-h^\vee}(\mathfrak{g})$	conformal	yes	excluded (Feigin–Frenkel center)	wild

The residual conjectural content of MC4 is thus: the $\mathrm{MC4}^\circ$ regime for class-`M` principal (Virasoro, principal $\mathcal{W}_N$) pending explicit chain-level SDR inscription and resolution of the Feigin–Frenkel chiral-coproduct obstruction of Proposition A.3; non-principal hook-type pending a chain-level SDR for the parabolic presentation; subregular / minimal $\mathcal{W}$ pending a theorem-level reduction to the class-C $\beta\gamma$ SDR (the present text supplies only a prose observation); and $V_{-h^\vee}(\mathfrak{g})$ (critical level), where the Feigin–Frenkel centre forces $\rho = \infty$. These are genuine remaining mathematics at the bar-complex chain level, not completion-machinery plumbing.

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